

Answers:

1. *Describe the difference between degrees and radians. When would you use one over the other in game development?*
 - a. Generally the idea here should be to understand that degrees are more intuitive, and used in everyday situations. In gamedev contexts, you'd find them in 3d modelling software and most interfaces that require interaction with users. Radians, on the other hand, are more natural for mathematical calculations, and would be used in code.
2. *What is a unit circle and why is it important in the discussion of degrees and radians?*
 - a. Unit circle is a circle of radius 1, centered at the origin. It's important generally because it provides an easy way to compute values such as sin/cos of angles, whether they're in degrees or radians.
3. *Why does a full circle have 2π radians?*
 - a. A full circle has 2π radians because of how radians are defined. Recall that 1 radian is 1 unit of length around the perimeter of the unit circle, and by definition, the circumference of a circle is $2\pi r$. Using a unit circle, $r=1$, thus the unit circle has 2π radians.
4. *How do radians relate to the circumference of a circle?*
 - a. They are directly related to the circumference of a circle. One radian is defined as the angle subtended when the arc length equals the radius.
5. *Explain the significance of π when working with radians.*
 - a. π relates the radius of a circle to its circumference with the formula $2\pi r$.
6. *Convert 90° to radians.*
 - a. $\pi/2$ radians
7. *Convert π radians to degrees.*
 - a. 180°
8. *Convert 45° to radians.*
 - a. $\pi/4$ radians
9. *Convert $\pi/2$ radians to degrees.*
 - a. 90°
10. *Convert $\pi/3$ radians to degrees.*
 - a. 60°
11. *Convert $\pi/4$ radians to degrees.*
 - a. 45°
12. *Convert $\pi/6$ radians to degrees.*
 - a. 30°

13. Convert 3 radians to degrees.

a. $3 * 180 / \pi = \sim 171.89^\circ$

14. Calculate the angle, in radians, between 2 vectors at 45° and 135° .

a. $135^\circ - 45^\circ = 90^\circ$

b. $90^\circ = \pi/2$ radians